Physical Layer: Transmission Media

Summarized Notes — Unit II (Computer Networks)

1. Transmission Media — Overview

A transmission medium is anything that can carry information from a source to a destination. In data communication it refers to free space, a metallic cable, or a fiber-optic cable through which a signal travels. Transmission media are technically located below the physical layer, and their use is directly controlled by the physical layer. The information transmitted is usually a signal — a converted form of the original data. In telecommunications, transmission media are broadly divided into two categories: **Guided (wired)** media, which provide a physical channel from one device to another (twisted-pair cable, coaxial cable, fiber-optic cable), and **Unguided (wireless)** media, which transport electromagnetic waves without a physical conductor (radio waves, microwaves, infrared).

2. Guided Media (Wired)

Guided media provide a conducting channel from one device to another, so the signal is directed and physically contained within the medium. Twisted-pair and coaxial cables use metallic (copper) conductors that carry signals as electric current, while optical fiber carries signals in the form of light.

2.1 Twisted-Pair Cable

A twisted pair consists of two insulated copper conductors twisted together. One wire carries the signal to the receiver while the other acts as a ground reference; the receiver reads the difference between the two. Noise and crosstalk affect both wires, but because the wires are twisted, both are exposed almost equally to external interference at every twist. This balancing effect cancels out most of the unwanted signal at the receiver — the more twists per unit length, the better the noise immunity.

Unshielded Twisted-Pair (UTP)

UTP is the most common twisted-pair cable used in communications. It has a plastic outer cover with no additional metallic shielding.

Shielded Twisted-Pair (STP)

STP, developed by IBM, adds a metal foil or braided mesh shield around each pair of insulated conductors. This shielding reduces noise and crosstalk penetration but makes the cable bulkier and more expensive than UTP.

Categories of UTP Cable

Category	Specification	Data Rate (Mbps)	Use
1	UTP used in telephone lines	< 0.1	Telephone
2	UTP originally used in T-lines	2	T-1 lines
3	Improved CAT 2, used in LANs	10	LANs
4	Improved CAT 3, used in Token Ring	20	LANs
5	24 AWG wire with jacket and sheath	100	LANs
5E	Extension of CAT 5; reduces crosstalk & EMI	125	LANs
6	Matched components from same manufacturer	200	LANs
7	SSTP; each pair individually foil-wrapped	600	LANs

Connectors

The most common UTP connector is the RJ45 (Registered Jack). It is a keyed connector, meaning it can only be inserted in one orientation.

Applications of Twisted-Pair Cable

Applications:

- Used in telephone lines to provide voice and data channels
- Forms the local loop (last-mile connection) linking subscribers to the central telephone office
- Used by telephone companies for DSL (Digital Subscriber Line) high-data-rate connections
- Used in LAN standards such as 10Base-T and 100Base-T

2.2 Coaxial Cable

Coaxial cable (coax) carries signals of higher frequency ranges than twisted-pair cable. Instead of two separate wires, it has a central core conductor (solid or stranded copper) surrounded by an insulating sheath (dielectric), which is in turn surrounded by an outer conductor (metal foil or braided shield), and finally an outer plastic cover. The outer conductor acts both as a shield against noise and as the second conductor completing the circuit, giving coax better shielding and higher bandwidth than twisted-pair cable.

Coaxial Cable Connectors

Coaxial cable is connected to devices using coaxial connectors. The most common type today is the Bayonet Neill-Concelman (BNC) connector, which uses a twist-lock mechanism. Three popular connector types are used:

- **BNC connector** — connects the end of the cable to a device such as a TV set
- **BNC T connector** — used in Ethernet networks to branch out to a computer or other device
- **BNC terminator** — placed at the end of the cable to absorb the signal and prevent reflection

Categories of Coaxial Cable

Category	Impedance	Use
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Applications of Coaxial Cable

Applications:

- Analog telephone networks — a single coaxial network could carry up to 10,000 voice signals
- Digital telephone networks — a single coaxial cable could carry digital data up to 600 Mbps
- Cable TV networks — the entire network originally used coaxial cable (Cable TV uses RG-59); later, providers replaced much of the media with fiber-optic cable
- Traditional Ethernet LANs — 10Base-2 (Thin Ethernet) uses RG-58 cable with BNC connectors (10 Mbps, up to 185 m); 10Base5 (Thick Ethernet) uses RG-11 cable (10 Mbps, up to 5000 m)

2.3 Fiber-Optic Cable

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light rather than electric current. Light travels in a straight line through a uniform substance; when it passes from a denser to a less dense substance, it bends. If the angle of incidence is less than the critical angle, the ray refracts; if equal to the critical angle, it bends along the interface; and if greater than the critical angle, the ray is reflected back into the denser medium. Optical fibers exploit this reflection principle to guide light: a glass or plastic core is surrounded by a cladding of less dense material, so light travelling through the core is continually reflected off the cladding rather than escaping through refraction.

Propagation Modes

Current technology supports two propagation modes for light in optical fiber — multimode and single mode — each requiring fiber of different physical construction. Multimode can further be implemented as step-index or graded-index fiber.

- **Multimode Step-Index** — the core has constant density; light travels straight until it hits the core–cladding interface, where it changes angle abruptly. This sudden 'step' change distorts the signal.
- **Multimode Graded-Index** — the core density is highest at the centre and gradually decreases toward the edge, which reduces signal distortion compared to step-index fiber.
- **Single Mode** — uses step-index fiber with a highly focused light source, limiting beams to a narrow range of angles close to horizontal. The critical angle is close to 90°, so all beams travel almost horizontally and arrive together, causing negligible distortion.

Fiber-Optic Cable Connectors

- **SC (Subscriber Channel) connector** — used for cable TV; uses a push/pull locking system
- **ST (Straight-Tip) connector** — used to connect cable to networking devices; uses a bayonet locking system and is more reliable than SC
- **MT-RJ connector** — the same size as an RJ45 connector

Applications of Fiber-Optic Cable

Applications:

- Backbone networks, due to its cost-effective wide bandwidth
- Wavelength-Division Multiplexing (WDM) systems, capable of data rates up to 1600 Gbps
- SONET networks
- Local-area networks such as 100Base-FX (Fast Ethernet) and 1000Base-X

Advantages:

- Higher bandwidth — supports dramatically higher data rates than twisted-pair or coaxial cable
- Less signal attenuation — a signal can travel up to 50 km without regeneration, versus every 5 km for copper cable
- Immunity to electromagnetic interference — noise cannot affect fiber-optic signals
- Resistance to corrosive materials — glass resists corrosion better than copper
- Light weight — much lighter than equivalent copper cable
- Greater immunity to tapping — copper cables create antenna effects that are easy to tap; fiber does not

Disadvantages:

- Installation and maintenance require specialized expertise not yet available everywhere
- Light propagation is unidirectional — two fibers are needed for bidirectional communication
- Cable and interface costs are higher than other guided media, which may not be justified for low-bandwidth needs

3. Unguided Media (Wireless)

Unguided media transport electromagnetic waves without using a physical conductor — this is commonly called wireless communication. Signals are broadcast through free space and are available to anyone with a receiving device. Wireless communication uses the part of the electromagnetic spectrum ranging from 3 kHz to 900 THz. Unguided signals travel from source to destination through three propagation methods:

- **Ground propagation** — low-frequency radio waves travel through the lowest part of the atmosphere, radiating in all directions and following the curvature of the earth. Greater transmission power gives greater range.
- **Sky propagation** — higher-frequency radio waves radiate upward into the ionosphere, where they are reflected back to earth, allowing long-distance communication with less power.
- **Line-of-sight propagation** — very high-frequency signals travel in a straight line directly from antenna to antenna, which must be directional, facing each other, and tall or close enough to avoid the earth's curvature.

The radio-wave and microwave portion of the spectrum is divided into eight government-regulated bands, from Very Low Frequency (VLF) to Extremely High Frequency (EHF):

Band	Range	Propagation	Application
VLF	3–30 kHz	Ground	Long-range radio navigation
LF	30–300 kHz	Ground	Radio beacons, navigational locators
MF	300 kHz–3 MHz	Sky	AM radio
HF	3–30 MHz	Sky	Citizens band (CB), ship/aircraft
VHF	30–300 MHz	Sky & Line-of-sight	VHF TV, FM radio
UHF	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular, paging, satellite
SHF	3–30 GHz	Line-of-sight	Satellite
EHF	30–300 GHz	Line-of-sight	Radar, satellite

3.1 Radio Waves

Electromagnetic waves ranging in frequency from 3 kHz to 1 GHz are called radio waves. Radio waves are omnidirectional — when an antenna transmits, the waves propagate in all directions, which makes them susceptible to interference from other antennas using the same frequency or band. Radio waves, especially those using sky-mode propagation, can travel very long distances. However, the radio-wave band is relatively narrow (just under 1 GHz), so subdividing it into subbands yields narrow channels and a comparatively low data rate for digital communication.

Omnidirectional Antenna

Radio waves use omnidirectional antennas that send signals in all directions. Depending on wavelength, strength, and purpose, several antenna types exist. This omnidirectional property makes radio waves useful for multicasting — one sender, many receivers — as seen in AM/FM radio, television, maritime radio, cordless phones, and paging.

3.2 Microwaves

Electromagnetic waves with frequencies between 1 GHz and 300 GHz are called microwaves. Unlike radio waves, microwaves are unidirectional — the sending and receiving antennas must be aligned with each other. This allows one pair of aligned antennas to operate without interfering with another pair.

Characteristics of Microwave Propagation

- Propagation is strictly line-of-sight, so towers that are far apart must be very tall to remain in direct sight of one another
- The earth's curvature and obstacles prevent short towers from communicating directly; repeaters are often required for long-distance links
- Very high-frequency microwaves cannot penetrate walls, which is a disadvantage for receivers located inside buildings
- The microwave band is wide (almost 299 GHz), allowing wide subbands and high data rates
- Use of certain portions of the microwave band requires permission from regulatory authorities

Unidirectional Antennas

Microwave communication uses two types of unidirectional antennas:

- **Parabolic dish antenna** — based on parabola geometry; every line parallel to the line of symmetry reflects off the curve and converges at a single focus point, so the dish acts as a funnel that catches a wide range of waves and concentrates them, recovering more signal than a single-point receiver. For transmission, the horn broadcasts toward the dish, which deflects the waves outward.
- **Horn antenna** — shaped like a giant scoop; outgoing signals travel up the stem and are deflected outward in narrow parallel beams by the curved head, while incoming signals are collected by the scoop shape and funneled down into the stem.

Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

3.3 Infrared Waves

Infrared waves have frequencies ranging from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm) and are used for short-range communication.

- Used for short-range communication within a closed area using line-of-sight propagation
- High-frequency infrared waves cannot penetrate solid walls
- Because they cannot pass through walls, an infrared system in one room does not interfere with a system in the next room
- Requires a direct line of sight between transmitter and receiver
- Common examples: TV remote controls, IR wireless headphones, IR data transfer between devices, motion detectors

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